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TWIRL repository audit and near-term execution plan

A reproducibility, production-safety, and scientific-readiness review of the standalone survey pipeline.

Abstract

We audited TWIRL as an executable survey pipeline, emphasizing reproducibility, production safety, validation semantics, generated-artifact hygiene, and alignment between the Stage 1-5 plan and the code that now exists. The repository has a credible pilot foundation: Stage 1 produces validated A2v1 light curves, Stage 2 supports an interpretable periodic search, and the fast validation suite is healthy. It is not yet a survey-complete system. The critical gaps are a frozen release manifest, science-level photometric QA, the non-periodic dip branch, multi-sector aggregation, false-alarm calibration, frozen-product injection recovery, and on-target end-to-end validation. Teacher development should remain bounded triage work while these baseline survey gates move onto the critical path.

Audit scope

Dimension	Evidence reviewed
Pipeline	Stage 1-5 implementation, interfaces, and acceptance gates
Validation	Fast tests, docs, detection sample, parsers, links, and syntax
Operations	Git branches/worktrees, artifact boundaries, local data, and production safety
Science	Claim boundaries, benchmark evidence, recovery semantics, and holdout use

Headline findings

1. A2v1 generation, validation, compact export, and periodic BLS now form a credible pilot path. **2.** Integrity-valid is not yet science-ready: Tier-1 photometric QA remains open. **3.** Teacher-v2 and early S57 labels are exploratory evidence and should not silently redefine the baseline or holdout. **4.** Git history is artifact-heavy; prospective curation is safer than history surgery during the active branch stack.

Bottom line: the next advance should be evidentiary rather than architectural.

Methods

The scan combined repository-wide inventory, Git/worktree inspection, parser and link checks, shell and Python syntax checks, the project fast test suite, and review of the authoritative plan, production protocol, data conventions, and plotting rules. The integrated checkpoint contained 2,137 tracked files and 759.6 MiB of tracked content; the fast suite completed with 286 passed and 3 skipped tests. We separated a Tier-0 integrity gate (schema, coverage, openability, and benchmark checks) from the still-open Tier-1 science QA gate (photometric precision, cadence loss, aperture outliers, injection preservation, and an independent extraction comparison).

Pipeline readiness

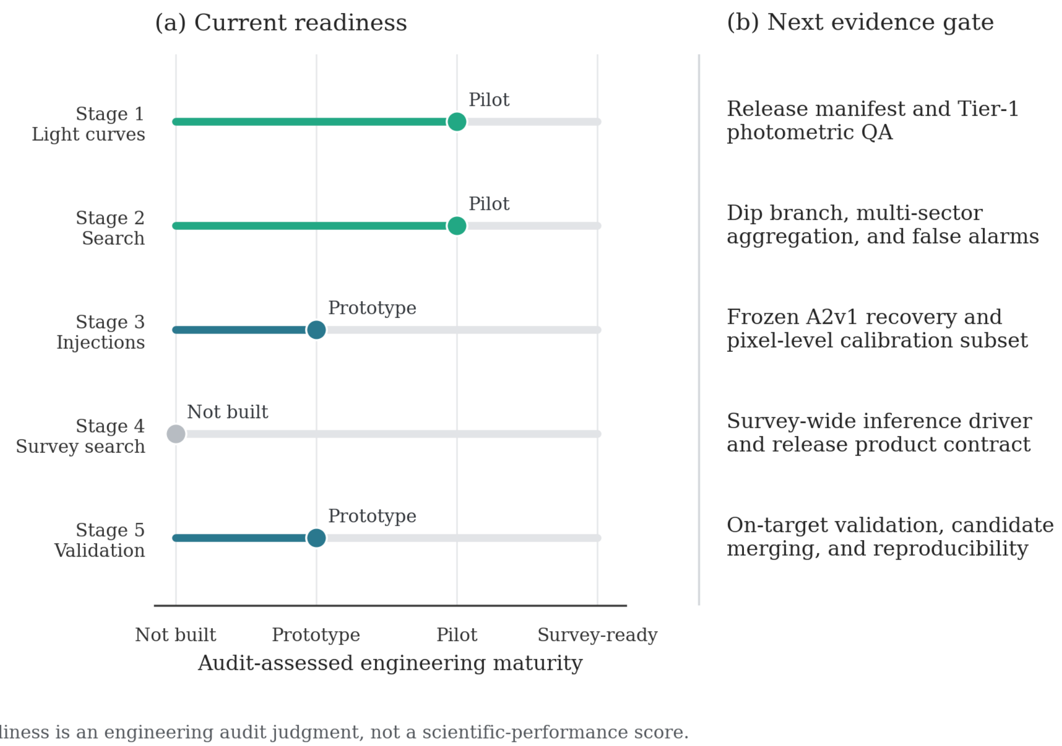

Figure 1. Audit-assessed engineering maturity and the next evidence gate for each pipeline stage. The scale describes implementation maturity, not scientific performance. Stage 1 and the periodic part of Stage 2 are at pilot maturity; Stage 3 and Stage 5 remain prototypes; Stage 4 is not yet built.

The strongest result is separation of the production contract from later analysis. The main weakness is that an integrity-valid sector can still lack the evidence required for scientific readiness. The clean forward plan also diverged from experimental teacher-v2 and S57 labeling work already present. Those products should be preserved with provenance but quarantined from the production baseline, and S57 must no longer be described as a pristine holdout.

Results

Repository hygiene is acceptable at the source level but weak at the artifact-history boundary. At scan time the Git object store was approximately 14 GiB and dominated by historical generated outputs. The on-disk reports tree was approximately 5.5 GiB because it also contained large ignored or untracked review artifacts, while source, scripts, and tests together occupied only about 15 MiB. Nonportable symlinks, stale caches, large third-party reference bundles, and redundant rendered material belong outside versioned survey products. The local environment reported NumPy 2.2.6 although pyproject.toml pins NumPy below 2.0; tests passed, so this is environment drift rather than an observed failure.

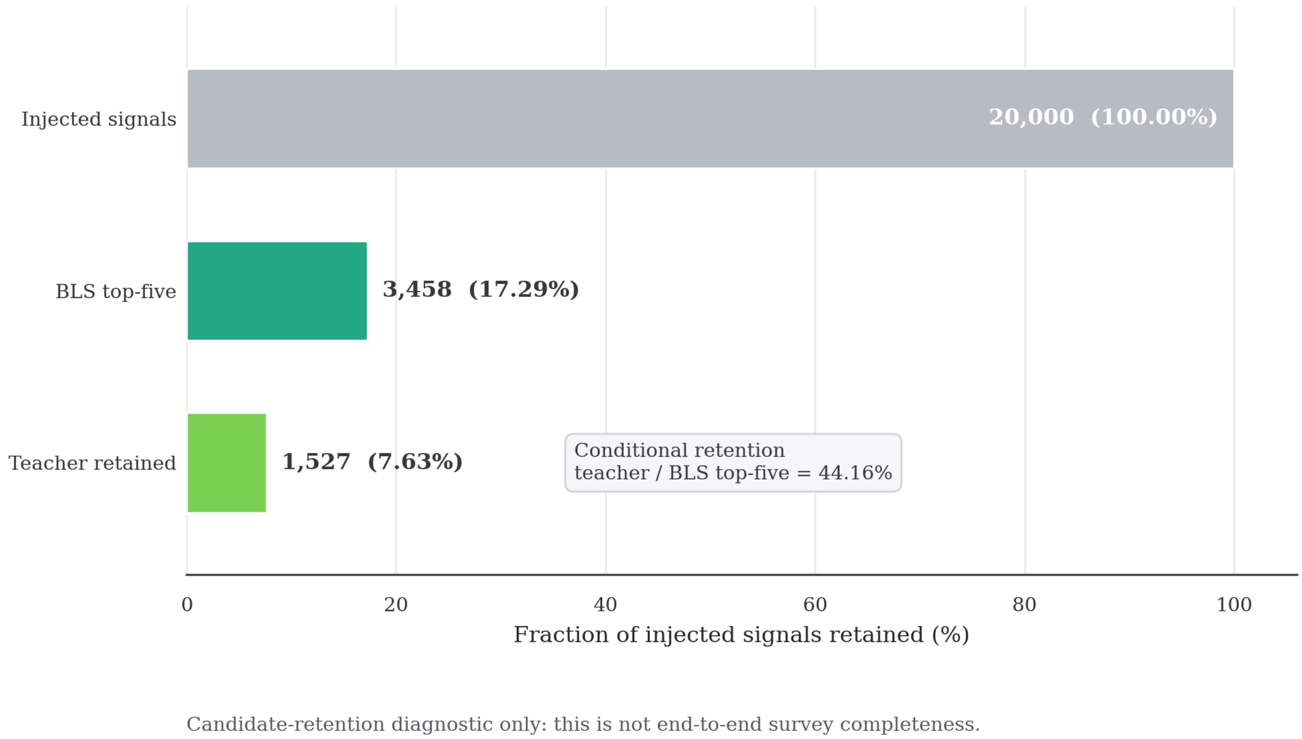

Figure 2. Candidate-retention diagnostic for the 20,000-signal teacher-v1 experiment. Top-five BLS retained 3,458 signals (17.29% of injections), and the teacher retained 1,527 (7.635% of injections; 44.16% conditional on top-five BLS recovery). This is not end-to-end survey completeness.

The attrition localizes losses, but it also shows why classifier iteration cannot substitute for baseline search and calibration. Reserve *end-to-end recovery* for a frozen chain that includes search, optional ranking, vetting, candidate merging, and the adopted calibration products. The present teacher result is a candidate-retention efficiency.

Period-radius candidate-retention surface

The aggregate retention counts localize pipeline attrition, while the canonical Teacher-v1 surface shows where the periodic search retains candidates across injected period, companion radius, and target brightness. It is included as diagnostic evidence, not as a final survey selection function.

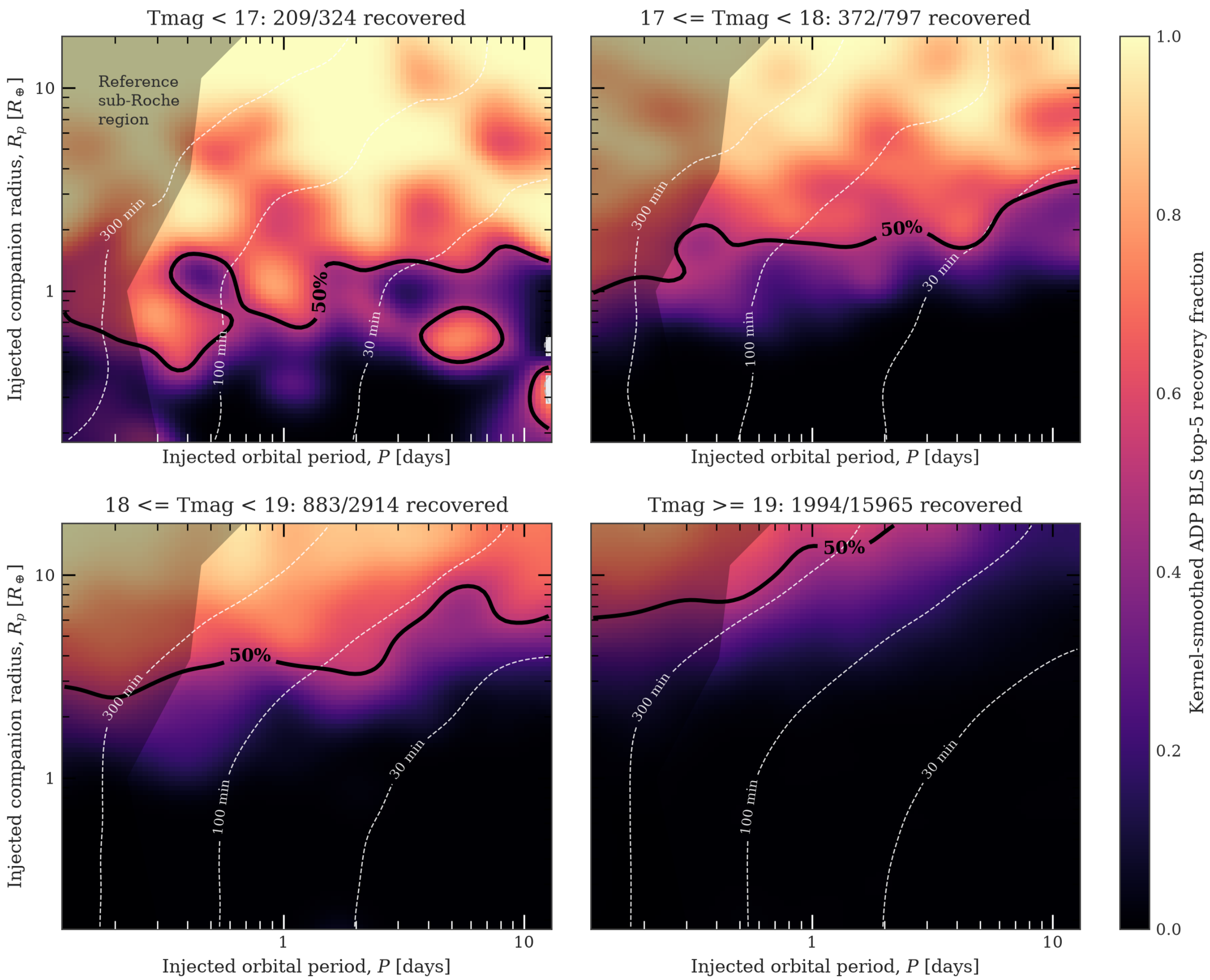

Figure 3. A2v1 Teacher-v1 BLS top-five candidate-retention surface for 20,000 injected signals, split into four Tmag strata. Fractions are kernel-smoothed; recovered and injected support counts are printed in every panel, and the bright bin is comparatively sparse. This is not end-to-end survey completeness and is not an occurrence-rate selection function.

A frozen survey analysis must propagate the same injections through the adopted vetter, candidate merger, and calibration chain before interpreting this surface as completeness.

Recommendations

1 | Freeze the release contract	Record a TWIRL-I sector cutoff, target/sample version, product tag, search configuration, and checksums in a machine-readable manifest.
2 | Complete Tier-1 science QA	Add precision-versus-magnitude regression, cadence-loss distributions, aperture thresholds, fixed injection preservation, and an independent benchmark extraction.
3 | Finish the transparent baseline	Implement the dip branch, multi-sector consolidation, and empirical false-alarm strategy before adding new ML gates.
4 | Calibrate frozen-product recovery	Run the adopted search, optional ranker, vetter, and merger; include a bounded pixel-level subset to capture extraction and detrending losses.
5 | Quarantine exploratory teacher work	Preserve teacher-v2 and early S57 labels with provenance, pause further holdout use, and require external-retention plus morphology checks before promotion.
6 | Control repository growth	Version compact summaries, manifests, selected figures, and small label tables; ignore rendered sheets, dependency bundles, shards, and local literature copies.
7 | Rebuild and automate the environment	Rebuild from pyproject.toml before release validation. After settling the dependency or lock strategy, add PR CI for make test-fast and make check-docs. This is medium priority below the science gates.

Execution order

Finish the A2v1 queue with Tier-0 gates and compact exports; freeze the release manifest and implement Tier-1 QA; complete dip, multi-sector, and false-alarm baselines; run frozen-product recovery and pixel calibration; then perform survey-wide inference and on-target validation. Bounded labeling may proceed in parallel only when it does not delay these gates.

Conclusion

TWIRL is beyond a collection of experiments: it has a testable pilot pipeline and a clear production contract. Freezing the release, separating integrity from science QA, completing the transparent search branches, and measuring recovery through the whole adopted chain will convert the pilot into a defensible survey framework.

Reproducibility

Figures and this PDF are generated by scripts/build_repository_audit_report.py. The source report is repository_audit_report.md. The project plan and progress log remain the authoritative status records.